COMET: Comic Multimodal Extraction and Translation

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Abstract

This project studies automated translation and re-typesetting of manga pages using a multimodal pipeline that combines computer vision, optical character recognition, and machine translation. Given a manga page image, the system detects speech bubbles, extracts the text inside them, translates the text into English, and inserts the translated text back into the original bubbles while preserving layout and readability. The goal is to evaluate how well current vision and language components can work together to support real-world document translation tasks that require both textual understanding and spatial reasoning.

Keywords

multimodal NLP, manga translation, OCR, machine translation, document layout

1 Project Topic and Number

Group 3: Multimodal and Cross-Modal NLP
Topic: *Multimodal Reasoning with Text and Vision*

2 Problem Statement

Many manga and comic books are not available in translated form because manual translation and re-typesetting is time-consuming and requires artistic effort. Although there are various machine translation tools, they do not handle visual layout, speech bubbles, or font sizing, which are critical for comics. Automatically translating manga requires joint reasoning on text content and visual structure. The main problem addressed in this project is how to reliably extract, translate, and reinsert text into manga speech bubbles while preserving the original page layout.

3 Proposed Technique

The proposed system is a modular, end-to-end pipeline that translates manga pages while preserving their original visual layout. The pipeline operates directly on page images and combines computer vision, optical character recognition, and machine translation with layout-aware text rendering.

Given an input manga page, the system first performs basic image preprocessing such as grayscale conversion and contrast normalization to improve downstream detection and OCR robustness. Speech bubbles are then detected using either lightweight image-based heuristics or a pretrained object detection model. Each detected bubble is processed independently to simplify layout reasoning and error analysis.

For each speech bubble, a segmentation step extracts the interior region of the bubble, excluding the outline. This interior mask defines the writable area available for text placement. Optical character recognition (OCR) is applied within the bubble region to extract the source text. The system also estimates text orientation to identify vertical Japanese text, which is common in manga, and records this information for analysis. While OCR accuracy for stylized vertical text is a known challenge, the extracted text is sufficient to support downstream translation and layout reconstruction.

The extracted text is translated into English using an existing machine translation model. Translation is performed at the bubble level, allowing the system to treat each speech unit independently. To reduce layout issues caused by length expansion during translation, the system applies simple length control strategies such as text truncation or rephrasing when needed.

To re-typeset the translated text, a layout-aware text fitting algorithm is used. This algorithm iteratively selects font size and line breaks so that the translated text fits entirely within the segmented bubble interior. Text placement is centered within the bubble, and rendering constraints are enforced to prevent overflow beyond the writable region. Before rendering, the original text inside the bubble is removed using region-based inpainting or interior repainting. The translated text is then rendered into the cleaned bubble and composited back into the original page image.

Each stage of the pipeline produces intermediate outputs, including detected bubble regions, segmentation masks, extracted text, translated text, and final rendered pages. These artifacts enable quantitative evaluation, visualization of failure cases, and systematic analysis of how errors propagate through the pipeline.

4 Expected Outcomes

The project is expected to produce a working prototype that translates and re-typesets a small set of manga pages while preserving visual layout. The system will output translated pages along with intermediate visualizations for each pipeline stage.

Evaluation will be based on:

- OCR confidence scores on extracted text,
- translation quality on a small reference set,
- layout correctness measured by text overflow rate and font size statistics.

Qualitative results will include before-and-after visual comparisons and analysis of failure cases such as stylized fonts or dense text regions. These results will demonstrate the feasibility of multimodal approaches for document translation tasks that require joint text and visual reasoning.

5 Scope and Feasibility

The project is designed to be feasible within a limited time budget by using existing OCR and translation models rather than training new ones. The initial scope focuses on a single language pair and horizontal English text rendering. More advanced features such as improved contrast normalization or extended language support are treated as optional extensions. With incremental development and evaluation on a small dataset, the project can be completed within the semester timeline.